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Upgrading the control electronics of an 8-inch Orion telescope
Inches

This project involved modernizing the electronics of an Orion telescope whose original controller had failed due to environmental wear. This telescope has two encoders per axis: a high-resolution encoder for trajectories and fine object tracking, and a low-resolution encoder for pointing. Since the motors and encoders were still in good condition, they were reused, and a new control system based on a modular architecture was implemented.

Two ESP32 microcontrollers were used, one for each axis (altitude and azimuth). These microcontrollers control the movements of each motor with the help of an L298N H-bridge and process the optical encoder signals in real time. This required designing an electronic conditioning stage for each encoder, which includes signal amplification using an LM324 in non-inverting configuration, offset correction with a potentiometer, and digitization using inverter gates.

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The conditioning circuit was tested on a protoboard and then transferred to a printed circuit board that was made in the same laboratory.

Each ESP32 calculates trajectories based on encoder pulses and transmits position and status information to a Raspberry Pi via serial communication. This Raspberry Pi acts as the system's central node, running a Python script that receives coordinates and allows commands to be sent to the system.

In addition, Stellarium software is being integrated, which will allow users to select celestial objects in real time. Stellarium will send the coordinates directly to the Raspberry Pi. This feature is in its final stages of integration and testing.

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